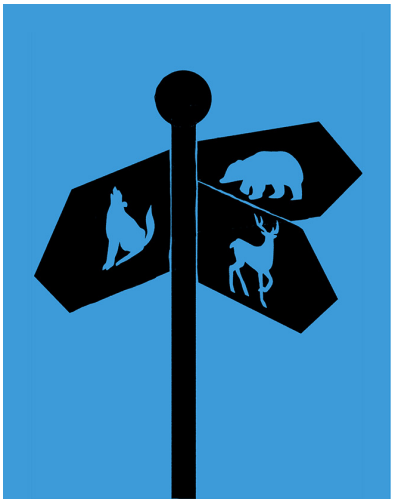


POPULATIONS ARE SUBJECTED TO UNPREDICTABLE FORCES

THE NEUTRAL THEORY OF BIODIVERSITY

IN CONTEXT

KEY FIGURES

Hal Caswell (1949–), **Stephen P. Hubbell** (1942–)

BEFORE

1920 Frederic Clements describes how plant species are associated with each other in communities.

1926 Henry Gleason proposes that ecological communities are organized more randomly.

1967 Richard Root introduces the concept of the ecological guild—a group of species exploiting resources in similar ways.

AFTER

2018 A review headed by Dutch ecologist Marten Scheffer suggests that, although species that use the same resources may be competitively equivalent, they may also differ according to their response to stress-inducing factors, such as drought or disease.

Biodiversity is shaped globally by new species appearing and others becoming extinct. Community ecology has traditionally held that interactions between species play a vital role in determining this process. If two species compete for similar resources, for example, either the stronger